

PHONEPE PULSE DATA VISUALIZATION AND EXPLORATION

Project Report

This report presents the design and implementation of a PhonePe Pulse Analytics Dashboard built using Power BI. The project leverages transaction and user datasets to analyze digital payment growth, customer engagement, regional performance, and market trends across India. The dashboard provides interactive visualizations and key performance indicators (KPIs) that help convert complex data into meaningful business insights.

1. Project Overview

- **Project Title:** PhonePe Pulse Analytics Dashboard
- **Domain:** FinTech / Digital Payments
- **Tool Used:** Microsoft Power BI
- **Data Source:** PhonePe Pulse GitHub Repository
- **Datasets Used:** Aggregated Transaction, Aggregated User, Top Transaction, Top User, Map Transaction, Map User
- **Dashboard Pages:** Transaction Analysis Dashboard & User Analysis Dashboard
- **Key KPIs:** Transaction Amount, Transaction Count, Registered Users, User Growth, Top States Analysis

2. Business Problem Statement

The PhonePe Pulse generates a large volume of transaction and user data across India. Analyzing this raw data manually makes it difficult to identify transaction trends, user adoption patterns, and regional performance. The objective of this project is to transform complex data into interactive dashboards that provide meaningful business insights. These insights help stakeholders make data-driven decisions and identify growth opportunities.

3. Business Objectives

- Analyze transaction and user growth trends across India.
- Identify top-performing states and districts.
- Understand customer payment behavior and preferences.
- Monitor key business KPIs through interactive dashboards.
- Generate actionable insights to support business decision-making and growth strategies

4. Data Collection and Preprocessing

The data used in this project was sourced from the PhonePe Pulse GitHub Repository, which contains transaction and user information across various states and districts in India. The datasets were imported into Power BI and prepared for analysis through data cleaning and transformation processes. This included validating data quality, standardizing column formats, handling inconsistencies, and creating calculated measures to ensure accurate reporting and visualization.

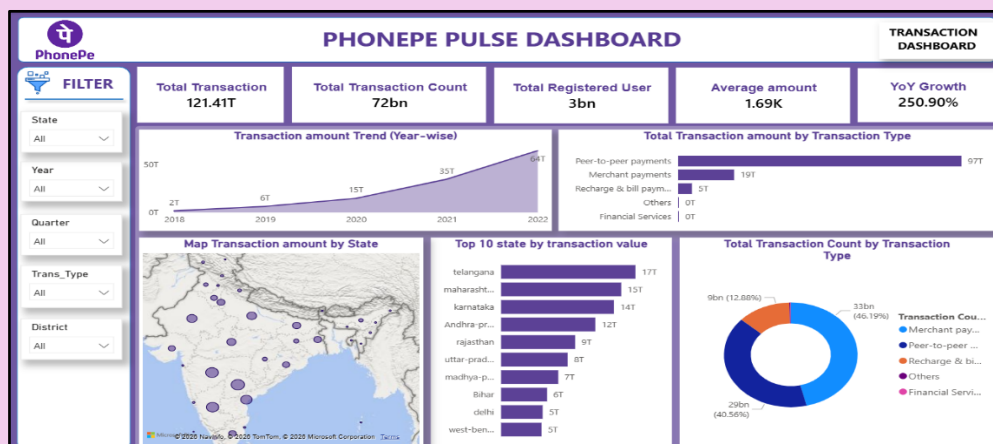
5. Tools and Technologies Used

- **Power BI Desktop** – Data visualization, dashboard development, and business reporting.
- **Microsoft Excel** – Data review, validation, and preliminary analysis.
- **Power Query** – Data cleaning, transformation, and preprocessing.
- **DAX** – Creation of KPIs, measures, and calculated metrics.
- **Map Visualization** – State-wise and district-wise geographical analysis.

6. Dashboard Features

- Interactive KPI cards
- Transaction and User Analysis Dashboards
- Geo-map visualization
- State and district wise analysis
- Brand-wise user analysis
- Year and quarter filtering
- Top states and top districts comparison
- YoY growth Tracking
- Dynamic slicers and filters
- Trend analysis and performance tracking
- Interactive charts and visual insight

7. Dashboard Screenshots:



38. Key Business Insights

- The transaction amount showed continuous growth from 2018 to 2022, indicating increasing digital payment adoption.
- Top states contributed significantly to total transaction value and user registrations.
- The geo-map visualization highlighted strong transaction activity across major states in India.
- Merchant Payments and Peer-to-Peer Payments were the most used transaction types.
- Brand-wise analysis showed that Xiaomi, Samsung, and Vivo had the highest user share.
- District-level analysis helped identify regions with high and low user engagement.
- User registrations increased steadily over the years, showing growth in the PhonePe platform.
- YoY growth KPIs indicated positive growth in both transaction and user activity.

9. Challenges Faced

- Cleaning and preprocessing data from multiple datasets.
- Standardizing state and district names for consistency.
- Handling missing and inconsistent data values.
- Creating KPI measures and DAX calculations for analysis.
- Designing interactive and user-friendly dashboards.
- Transforming raw data into meaningful business insights

10. Conclusion

The PhonePe Pulse Analytics Dashboard successfully transformed raw transaction and user data into meaningful business insights using Power BI. The analysis helped identify transaction trends, user growth patterns, top-performing states, and customer payment behavior across India. Through interactive visualizations and KPI tracking, the dashboard provides a clear understanding of digital payment adoption and regional performance. The insights generated from this project can support data-driven decision-making, business growth, and strategic planning.

11. Future Enhancements

- Integration of real-time data for live dashboard monitoring.
- Implementation of predictive analytics for future trend forecasting.
- Advanced district-level and regional analysis.
- Addition of customer segmentation and behavioral insights.
- Enhancement of KPI tracking and business performance metrics.
- Mobile-responsive dashboard development for better accessibility

12. References

- PhonePe Pulse Data Repository
- Microsoft Power BI Documentation
- Power Query Documentation
- DAX Documentation
- Power BI Community Resources
- Online Learning and Visualization Resources